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IMPLEMENT-ON-GROUND DETECTION USING VIBRATION SIGNALS

Abstract

Described herein are systems, methods, and other techniques for determining a period during which an implement of a construction machine is interacting with a ground surface. A vibration signal that is indicative of a movement of the implement is captured. One or more features are extracted from the vibration signal. The one or more features are provided to a machine-learning model to generate a model output. An implement-on-ground (IOG) start time and an IOG end time are predicted based on the model output, the IOG start time and the IOG end time forming the period.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] This application claims priority to and is a continuation of U.S. patent application Ser. No. 17/493,793, filed Oct. 4, 2021, and entitled “IMPLEMENT-ON-GROUND DETECTION USING VIBRATION SIGNALS,” the content of which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND OF THE INVENTION

[0002] Modern mobile machinery, including construction and agricultural machines, have dramatically increased the efficiency of performing various work-related tasks. For example, earthmoving machines employing automatic grade control systems are able to grade project areas using fewer passes than what was previously done manually. As another example, modern asphalt pavers and other road makers have allowed replacement of old roads and construction of new roads to occur on the order of hours and days instead of what once took place over weeks and months. Due to the automation of various aspects, construction and agriculture projects can be carried out by crews with fewer individuals than what was previously required. Much of the technological advances of mobile machinery are owed in part to the availability of accurate sensors that allow real-time monitoring of the condition and position of a machine's components and/or the environment surrounding the machine.

SUMMARY OF THE INVENTION

[0003] A summary of the various embodiments of the invention is provided below as a list of examples. As used below, any reference to a series of examples is to be understood as a reference to each of those examples disjunctively (e.g., “Examples 1-4” is to be understood as “Examples 1, 2, 3, or 4”).

[0004] Example 1 is a method of determining a period during which an implement of a construction machine is interacting with a ground surface, the method comprising: capturing a vibration signal that is indicative of a movement of the implement; extracting one or more features from the vibration signal; providing the one or more features to a machine-learning model to generate a model output; and predicting an implement-on-ground (IOG) start time and an IOG end time based on the model output, the IOG start time and the IOG end time forming the period during which the implement is interacting with the ground surface.

[0005] Example 2 is the method of example(s) 1, wherein the vibration signal is captured using a vibration sensor mounted to the construction machine.

[0006] Example 3 is the method of example(s) 2, wherein the vibration sensor includes an accelerometer and the vibration signal includes an acceleration signal.

[0007] Example 4 is the method of example(s) 2, wherein the vibration sensor includes a gyroscope and the vibration signal includes a rotation signal.

[0008] Example 5 is the method of example(s) 2, wherein the vibration sensor is mounted to the implement.

[0009] Example 6 is the method of example(s) 1-5, wherein the model output includes a set of IOG candidates corresponding to times at which the implement is interacting with the ground surface.

[0010] Example 7 is the method of example(s) 1-6, wherein the model output includes a set of implement-in-air (IIA) candidates corresponding to times at which the implement is not interacting with the ground surface.

[0011] Example 8 is the method of example(s) 1-7, wherein the one or more features include at least one of signal amplitude features or signal frequency features.

[0012] Example 9 is the method of example(s) 1-8, wherein the machine-learning model is a pre-trained support-vector machine.

[0013] Example 10 is the method of example(s) 1-9, further comprising: adjusting a ground surface map based on the period during which the implement is interacting with the ground surface and a path of the implement.

[0014] Example 11 is a system comprising: one or more processors; and a computer-readable medium comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform operations comprising: capturing a vibration signal that is indicative of a movement of the implement; extracting one or more features from the vibration signal; providing the one or more features to a machine-learning model to generate a model output; and predicting an implement-on-ground (IOG) start time and an IOG end time based on the model output, the IOG start time and the IOG end time forming the period during which the implement is interacting with the ground surface.

[0015] Example 12 is the system of example(s) 11, wherein the vibration signal is captured using a vibration sensor mounted to the construction machine.

[0016] Example 13 is the system of example(s) 12, wherein the vibration sensor includes an accelerometer and the vibration signal includes an acceleration signal.

[0017] Example 14 is the system of example(s) 12, wherein the vibration sensor is mounted to the implement.

[0018] Example 15 is the system of example(s) 11-14, wherein the model output includes a set of IOG candidates corresponding to times at which the implement is interacting with the ground surface.

[0019] Example 16 is the system of example(s) 11-15, wherein the model output includes a set of implement-in-air (IIA) candidates corresponding to times at which the implement is not interacting with the ground surface.

[0020] Example 17 is the system of example(s) 11-16, wherein the one or more features include at least one of signal amplitude features or signal frequency features.

[0021] Example 18 is the system of example(s) 11-17, wherein the machine-learning model is a pre-trained support-vector machine.

[0022] Example 19 is the system of example(s) 11-18, wherein the IOG start time and the IOG end time are predicted further based on a set of implement positions.

[0023] Example 20 is a method of determining a period during which an implement of a construction machine interacts with a ground surface, the method comprising: capturing a vibration signal that is indicative of a movement of the implement; determining a set of implement-on-ground (IOG) candidates corresponding to times at which the implement is interacting with the ground surface; determining a set of implement-in-air (IIA) candidates corresponding to times at which the implement is not interacting with the ground surface; and predicting an IOG start time and an IOG end time based on the set of IOG candidates and the set of IIA candidates, the IOG start time and the IOG end time forming the period during which the implement is interacting with the ground surface.

[0024] Example 21 is a non-transitory computer-readable medium comprising instructions that, when executed by one or more processors, cause the one or more processors to perform any of the methods of example(s) 1-10 or 20.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The accompanying drawings, which are included to provide a further understanding of the disclosure, are incorporated in and constitute a part of this specification, illustrate embodiments of the disclosure and together with the detailed description serve to explain the principles of the disclosure. No attempt is made to show structural details of the disclosure in more detail than may be necessary for a fundamental understanding of the disclosure and various ways in which it may be practiced.

[0026] FIGS. **1A-1C** illustrate an example modification of a ground surface map based on an implement path.

[0027] FIG. **2** illustrates an example prediction of an IOG period based on a vibration signal.

[0028] FIG. **3** illustrates an example implementation of one or more techniques of the present disclosure within a construction environment.

[0029] FIG. **4** illustrates an example machine control system.

[0030] FIG. **5** illustrates an example IOG detector.

[0031] FIGS. **6A** and **6B** illustrate examples of a vibration signal and its corresponding extracted frequency features, respectively.

[0032] FIGS. **7A** and **7B** illustrate examples of implement motion features.

[0033] FIG. **8** illustrates an example of a prediction of IOG periods based on a vibration signal and its corresponding amplitude features.

[0034] FIG. **9** illustrates an example of a prediction of an IOG end time based on a vibration signal and an implement distance.

[0035] FIG. **10** illustrates an example method.

[0036] FIG. **11** illustrates an example computer system comprising various hardware elements.

[0037] In the appended figures, similar components and/or features may have the same numerical reference label. Further, various components of the same type may be distinguished by following the reference label with a letter or by following the reference label with a dash followed by a second numerical reference label that distinguishes among the similar components and/or features. If only the first numerical reference label is used in the specification, the description is applicable to any one of the similar components and/or features having the same first numerical reference label, irrespective of the suffix.

DETAILED DESCRIPTION OF THE INVENTION

[0038] In some instances, a construction machine such as an earthmoving machine may maintain a map of the surrounding surface of the ground. This map, which may alternatively be referred to as a “ground surface map”, may include a current elevation or height for each of multiple locations surrounding the machine. As the construction machine carries out a construction project by lowering the implement, cutting into the ground surface, and removing and lifting earth, the ground surface map may be updated based on the path of the implement and the period of time during which the implement was interacting with the ground surface. For example, if the implement is cutting into the ground surface during a period of three seconds, then it can be expected that the new ground surface is roughly equal to the implement's position (e.g., the position of the tip of an excavator's blade) during those three seconds. As such, the ground surface map can be adjusted to reflect the new ground surface.

[0039] An accurate adjustment of the ground surface map is dependent on two factors, the first being a knowledge of the implement's position and path, and the second being a determination of the time period during which the implement is actually interacting with (e.g., digging into) the ground, which may alternatively be referred to as the “interaction period”, the “implement-on-ground (IOG) period”, or the “IOG instance”. If either of these factors cannot be determined accurately, the ground surface map may be unreliable and can include error spikes, particularly at the positions where the path of the implement intersects the original ground surface. For example,

if the IOG period is predicted to start before the actual dig start time, this may cause the ground level map to include a spike in the upward direction as the implement will be above the actual ground surface. Similarly, if the IOG period is predicted to end after the actual dig end time, this may also cause the ground level map to include a spike in the upward direction, as the implement will be finished digging and be above the actual ground surface.

[0040] Embodiments of the present disclosure provide techniques for accurately predicting the period during which the implement of a construction machine is interacting with the ground. Embodiments may include predicting an IOG start time at which the implement begins interacting with the ground and an IOG end time at which the implement has finished interacting with the ground, which may correspond to the beginning and end of the IOG period, respectively. The disclosed techniques may rely on a vibration signal captured using a vibration sensor mounted to the construction machine. The vibration signal may be indicative of the movement of the implement during a digging event. In various embodiments, the vibration sensor may be mounted directly to the implement or mounted elsewhere on the construction machine.

[0041] Some embodiments may utilize trained machine-learning models, such as neural networks, that receive features of the vibration signal as input to generate the IOG period. In some instances, a machine-learning model may generate a set of IOG candidates corresponding to times that it is predicted that the implement is interacting with the ground and a set of implement-in-air (IIA) candidates corresponding to times that it is predicted that the implement is not interacting with the ground. The sets of IOG and IIA candidates may be analyzed to identify clusters and to predict the IOG period, with the IOG period generally including as many IOG candidates as possible and excluding as many IIA candidates as possible.

[0042] Several advantages are achieved by way of the present disclosure. For example, as noted above, by predicting accurate IOG periods, embodiments of the present disclosure allow accurate adjustments of the ground surface map. Such adjustments may be made in real time as the construction machine is operating within a construction environment. As another example, embodiments provide the ability to track the movement of material (e.g., where the material comes from and where it goes to). In certain applications, such as mining, it may be desirable to know whether material is valuable ore or low-value spoil. This determination can be derived from accurate knowledge of when and where the implement contacted the ground surface.

[0043] In the following description, various examples will be described. For purposes of explanation, specific configurations and details are set forth in order to provide a thorough understanding of the examples. However, it will also be apparent to one skilled in the art that the example may be practiced without the specific details. Furthermore, well-known features may be omitted or simplified in order not to obscure the embodiments being described.

[0044] FIGS. **1A-1C** illustrate an example modification of a ground surface map **124** based on an implement path **114**, in accordance with some embodiments of the present disclosure. In FIG. **1A**, a construction machine **150** begins moving an implement **121** along implement path **114**. In various embodiments, implement path **114** may be set by an operator of the construction machine and/or may be automatically generated based on the details of a construction project or other work-related task. In FIG. **1B**, implement **121** follows implement path **114** and cuts into ground surface **112**, thereby modifying ground surface **112** by removing the earth above implement path **114** while leaving the earth below implement path **114**.

[0045] In FIG. **1C**, ground surface map **124** is adjusted based on a ground surface map adjustment **125**, which is determined based on implement path **114** and an IOG period corresponding to the time frame of the digging action. In the illustrated example, a previous ground surface map **124-1** is adjusted to produce a new ground surface map **124-2** by replacing previous ground surface map **124-1** with ground surface map adjustment **125** for the locations that ground surface map adjustment **125** includes data (e.g., elevations or heights). For the locations outside of where ground surface map adjustment **125** includes data, new ground surface map **124-2** is set equal to

previous ground surface map **124-1**.

[0046] FIG. **2** illustrates an example prediction of an IOG period **240** based on a vibration signal **242**, in accordance with some embodiments of the present disclosure. In the illustrated example, vibration signal **242** is captured using a vibration sensor while implement **221** is following an implement path **214** that passes through ground surface **212**. While not shown, the vibration sensor may be mounted directly to implement **221** or elsewhere on the construction machine. The illustrated vibration signal is an acceleration signal along a particular axis (or any direction in space) that includes both positive and negative acceleration values.

[0047] Based on the features in vibration signal **242**, an IOG start time **236** (between 1.3 minutes and 1.31 minutes) at which implement **221** begins digging along ground surface **212** and an IOG end time **238** (between 1.46 minutes and 1.47 minutes) at which implement **221** ends digging along ground surface **212** are predicted. In the illustrated example, vibration signal **242** is shown to have higher amplitudes during IOG period **240** compared to outside IOG period **240**. Also shown in FIG. **2** is a truth reference showing the actual IOG period (or IOG truth **239**) and the corresponding actual IOG start and end times, which illustrate the accuracy of the described techniques of using the vibration signature to predict IOG period **240**.

[0048] FIG. **3** illustrates an example implementation of one or more techniques of the present disclosure within a construction environment. Specifically, FIG. **3** shows a construction machine **350** being deployed at a construction site **310** and having the control thereof at least partially implemented by a control unit **360** which, in various embodiments, may be communicatively coupled to a position sensor **358** and a vibration sensor **365** through a wired and/or wireless connection. While construction site **310** is generally described herein as corresponding to an earthmoving site, the present disclosure may be applicable to a wide variety of construction, maintenance, or agricultural projects in which heavy equipment or mobile machinery are used. Similarly, while construction machine **350** is generally described herein as corresponding to an earthmoving construction machine, the various techniques described herein may be applicable to a wide variety of construction machines or heavy equipment such as graders, excavators, bulldozers, backhoes, pavers (e.g., concrete, asphalt, slipform, vibratory, etc.), compactors, scrapers, loaders, material handlers, combine harvesters, spreaders, and the like.

[0049] In some embodiments, construction machine **350** may include a tractor with wheels, axles, and a gasoline-, diesel-, electric-, or steam-powered engine for providing power and traction to construction machine **350** to drive along a desired path, often at a constant speed. Construction machine **350** may be a tracked vehicle that incorporates a continuous track of treads or track plates that are driven by the vehicle's wheels. An operator of construction machine **350** may provide inputs to control unit **360** using various input devices such as levers, switches, buttons, pedals, steering wheels, and touch screens, which can cause various actuators to move construction machine **350**.

[0050] In some instances, construction machine **350** may include an implement **321**, which may be the primary component of construction machine **350** that is controlled to interact with elements of construction site **310**. For example, at an earthmoving site, implement **321** may be the component of construction machine **350** that interacts with (e.g., pushes, scoops, cuts, etc.) the earth, such as the blade of a bulldozer, the bucket of an excavator, or the drum of a compactor. As another example, at an agricultural site, implement **321** may be the header of a combine harvester or the boom of a spreader. As another example, at a road construction site, implement **321** may be the screed of an asphalt paver.

[0051] It should be appreciated that, in some embodiments, implement **321** and construction machine **350** may be considered to be separate bodies that have a semi-rigid coupling between them. The coupling is semi-rigid in that the bodies can move relative to each other but they can also be fixed at a given orientation. Some examples of semi-rigid couplings used on construction or earthmoving equipment include c-frames, angle c-frames, push arms, L-shaped push arms, and the

like.

[0052] In some embodiments, control unit **360** may determine a geospatial position **322** of construction machine **350** based on sensor data captured by one or more sensors (e.g., position sensor **358**) mounted to construction machine **350**. For example, position sensor **358** may be a global navigation satellite systems (GNSS) receiver that receives wireless signals from one or more GNSS satellites **302**. By processing the received wireless signals, a position of the GNSS receiver may be calculated. During operation of construction machine **350**, geospatial position **322** is determined and is used to query a site design, which may provide translations between two-dimensional (2D) positions within construction site **110** and a desired ground surface or digging depth.

[0053] As described herein, in some embodiments, vibration sensor **365** may be used to capture a vibration signal **342** that is indicative of the movement of implement **321**. Vibration signal **342** may be provided to an IOG detector **334**, which may extract features from vibration signal **342** and provide the extracted features to a machine-learning model that is used to predict one or more IOG period(s) **340**. Each of IOG period(s) **340** may occur during a larger digging event (or digging period) and may each correspond to a time period during which implement **321** is interacting with the ground at construction site **310**. IOG period(s) **340** may be associated with IOG start time(s) **336** and IOG end time(s) **338**.

[0054] FIG. **4** illustrates an example machine control system **400**, in accordance with some embodiments of the present disclosure. Machine control system **400** may include various input devices **452**, sensors **454**, actuators **456**, and computing devices for allowing one or more operators of the construction machine to complete a work-related task. The components of machine control system **400** may be mounted to or integrated with the components of the construction machine such that the construction machine may be considered to include machine control system **400**. The components of machine control system **400** may be communicatively coupled to each other via one or more wired and/or wireless connections.

[0055] Machine control system **400** may include a control unit **460** that receives data from the various sensors and inputs and generates commands that are sent to the various actuators and output devices. In the illustrated example, control unit **460** receives input data **453** from input device(s) **452** and sensor data **455** from sensor(s) **454**, and generates control signal(s) **457** which are sent to actuator(s) **456**. Control unit **460** may include one or more processors and an associated memory. In some embodiments, control unit **460** may be communicatively coupled to an external computing system **462** located external to machine control system **400** and the construction machine. External computing system **462** may send instructions to control unit **460** of the details of a work-related task. External computing system **462** may also send alerts and other general information to control unit **460**, such as traffic conditions, weather conditions, the locations and status of material transfer vehicles, and the like.

[0056] In some embodiments, input device(s) **452** may receive input data **453** that indicates a desired movement of the vehicle, a desired movement of the implement, a desired height of the implement, an activation of one or more mechanisms on the implement (e.g., rotating the bucket of an excavator), and the like. Input device(s) **452** may include a keyboard, a touchscreen, a touchpad, a switch, a lever, a button, a steering wheel, an acceleration pedal, a brake pedal, and the like. In some embodiments, input device(s) **452** may be mounted to any physical part of the vehicle, such as within the cab of the vehicle.

[0057] In some embodiments, sensor(s) **454** may include one or more position sensor(s) **458** and/or vibration sensor(s) **465**. Position sensor(s) **458** may be a combination of GNSS receivers, which determine position using wireless signals received from satellites, and total stations, which determine position by combining distance, vertical angle, and horizontal angle measurements. Vibration sensor(s) **465** may include one or more sensors that detect movement of the components of the construction machine to which they are rigidly attached. In some instances, vibration

sensor(s) **465** may include one or more microphones or other acoustic sensors. In some instances, vibration sensor(s) **465** may include one or more inertial measurement unit (IMU) sensors and/or the elements thereof. For example, vibration sensor(s) **465** may include one or more gyroscopes for detecting angular acceleration, angular rate and/or angular position (or other rotational signals or data), one or more accelerometers for detecting linear acceleration, linear velocity, and/or linear position, and/or one or more magnetometers for detecting the above-listed types of data, among other possibilities.

[0058] In some embodiments (such as, for example, when implemented as a gyroscope), vibration sensor(s) **465** may directly detect angular rate and may integrate to obtain angular position, or alternatively a vibration sensor may directly measure angular position and may determine a change in angular position (e.g., compute the derivative) to obtain angular rate. In many instances, vibration sensor(s) **465** can be used to determine the yaw angle (rotation angle with respect to a vertical axis), the pitch angle (rotation angle with respect to a transverse axis), and/or the roll angle (rotation angle with respect to a longitudinal axis) of the construction machine.

[0059] Control unit **460** may include various controllers and modules to assist in the generation of control signal(s) **457** and the computation of other data. Each of the controllers and modules may include dedicated hardware and/or may be performed using the main processor and/or memory of control unit **460**. In some embodiments, control unit **460** may include a machine component position estimator **428** that estimates the positions of the machine's components based on sensor data **455**. For example, machine component position estimator **428** may use sensor data **455** to estimate the positions of the implement during an IOG instance. Optionally, machine component position estimator **428** may generate a model of the entire construction machine as it is currently situated within the construction site. In such embodiments, the model may be generated not only based on sensor data **455** but additionally using feedback from the actuator(s) **456** and/or by tracking control signal(s) **457**.

[0060] In some embodiments, control unit **460** may include an IOG detector **434** that uses vibration signal **442** along with other sensor data **455** to predict each IOG period that occurs during a digging period. Each IOG period may be fed into a ground surface mapper **425** that maintains a ground surface map. Ground surface mapper **425** may adjust the ground surface map based on each IOG period provided by IOG detector **434** along with the implement positions provided by machine component position estimator **428**.

[0061] Control signal(s) **457** may include direct current (DC) or alternating current (AC) voltage signals, DC or AC current signals, and/or information-containing signals. An example of an information-containing signal may be controller area network (CAN) message that may be sent along a CAN bus or other communication medium. In some instances, control signal(s) **457** include a pneumatic or hydraulic pressure. Upon receiving control signal(s) **457**, actuator(s) **456** may be caused to move in a specified manner, such as by extending, retracting, rotating, lifting, or lowering by a specified amount. Actuator(s) **456** may use various forms of power to provide movement to the components of the construction machine. For example, actuator(s) **456** may be electric, hydraulic, pneumatic, mechanical, or thermal, among other possibilities.

[0062] FIG. 5 illustrates an example IOG detector **534**, in accordance with some embodiments of the present disclosure. As inputs, IOG detector **534** may be provided with a vibration signal **542** and a set of implement position(s) **529**, which may be estimated by a machine component position estimator **528** based on sensor data **555** (and optionally based on vibration signal **542**). IOG detector **435** may include a feature extractor **546** that extracts one or more features **547** from vibration signal **542** (and optionally implement position(s) **529**).

[0063] In some embodiments, features **547** may include signal amplitude features **547-1** related to the amplitude of vibration signal **542**. For example, signal amplitude features **547-1** may include the time-varying amplitude of vibration signal **542**, the magnitude of the amplitude of vibration signal **542**, the average of the amplitude of vibration signal **542**, the standard deviation of the

amplitude of vibration signal **542**, the short-time average amplitude over long-time average amplitude (STA/LTA) of vibration signal **542**, the long-time vibration changes in vibration signal **542** (for example, when vibration signal **542** is captured by a gyro), among other possibilities. [0064] In some embodiments, features **547** may include signal frequency features **547-2** related to the frequency of vibration signal **542**. For example, signal frequency features **547-2** may include the time-varying frequency of vibration signal **542** (e.g., over sequential time windows), the high-frequency energy of vibration signal **542**, the low-frequency energy of vibration signal **542**, the power spectral density of vibration signal **542**, among other possibilities.

[0065] In some embodiments, features **547** may include implement motion features **547-3** determined based on implement position(s) **529** (and optionally based on vibration signal **542** when vibration signal **542** includes an IMU signal). For example, implement motion features **547-3** may include the curl or rotation of the implement (e.g., the curl of the excavator's bucket), the distance between the implement and a reference point on the construction machine (e.g., the distance between the implement and the cab), the direction of movement of the implement, the slewing of the construction machine, among other possibilities.

[0066] Features **547** may be provided to a machine-learning model **532** that generates an output based on features **547**. For example, features **547** may be used for the input to machine-learning model **532** and may be passed through the layers or nodes of machine-learning model **532**. The output generated by machine-learning model **532** may be referred to as a model output. The model output may include a set of IOG candidates **544** and/or a set of IIA candidates **545**. In various embodiments, IOG candidates **544** and IIA candidates **545** may be directly outputted by machine-learning model **532** or they may be obtained by performing one or more post-processing steps on the model output.

[0067] In some embodiments, IOG detector **534** may include a post-processing stage **551** that analyzes the model output (optionally including IOG candidates **544** and/or IIA candidates **545**) and predicts IOG period **540** and/or IOG start time **536** and IOG end time **538** based on the analysis. For example, post-processing stage **551** may include steps for identifying clusters of IOG candidates **544** and/or IIA candidates **545** and predicting IOG period **540** and/or IOG start time **536** and IOG end time **538** based on the identified clusters. In one example, the time window that encompasses a cluster of IOG candidates **544** may be used for the prediction of IOG period **540**. In another example, the time window between two clusters of IIA candidates **545** that excludes each may be used for the prediction of IOG period **540**. In another example, the transition between a cluster of IIA candidates **545** and a cluster of IOG candidates **544** may be used for the prediction of IOG start time **536** and the transition between a cluster of IOG candidates **544** and a cluster of IIA candidates **545** may be used for the prediction of IOG end time **538**. In some embodiments, machine-learning model **532** may directly output IOG period **540** and/or IOG start time **536** and IOG end time **538** without the use of post-processing stage **551**.

[0068] In some embodiments, post-processing stage **551** may use implement motion features **547-3** to predict IOG period **540** and/or IOG start time **536** and IOG end time **538**. For example, the movement of the implement towards the cab of the construction machine may be indicative of an IOG instance. In one example, the distance between the implement and the cab may be analyzed to determine a minimum distance value and its associated time, which may be a strong indicator of when IOG end time **538** may occur. In another example, a maximum distance between the implement and the cab and its associated time may be determined, which may be a strong indicator of when IOG start time **536** may occur.

[0069] Machine-learning model **532** may be a support-vector machine, a neural network, a convolutional neural network, a Bayesian network, among other possibilities. Machine-learning model **532** may be trained through a training process in which multiple training vibration signals are provided to feature extractor **546** to produce training features that are provided to machine-learning model **532**. For each iteration during the training process, IOG period **540** produced using

a training vibration signal may be compared to a truth reference to determine any difference (alternatively referred to as the “loss”) between the training result and the truth reference. The loss may be used to modify machine-learning model **532** (e.g., by modifying the weights or parameters of machine-learning model **532**) so as to reduce the loss during subsequent inferences using machine-learning model **532**. It should be noted that while machine-learning model **532** is illustrated as a single machine-learning model, multiple machine-learning models may be employed to handle different types of features **547** or different types of model outputs.

[0070] FIGS. **6A** and **6B** illustrate examples of a vibration signal **642** and its corresponding extracted frequency features **647**, respectively, in accordance with some embodiments of the present disclosure. In FIG. **6A**, an acceleration signal captured by an accelerometer is shown along with the truth reference indicating when the IOG periods occur. In FIG. **6B**, the power spectral density of the acceleration signal is shown along with the truth reference indicating when the IOG periods occur. In some instances, the illustrated vibration signal and frequency features may be used to train the machine-learning model by generating predicted IOG periods using the machine-learning model, comparing the predicted IOG periods to the truth reference, and modifying the machine-learning model based on the comparison.

[0071] FIGS. **7A** and **7B** illustrate examples of implement motion features, in accordance with some embodiments of the present disclosure. In FIG. **7A**, the implement distance between the implement and the cab (or bucket distance) is shown along with the truth reference indicating when the IOG periods occur. In FIG. **7B**, the implement rotation (or bucket rotation, i.e., the angle of the bucket) is shown along with the truth reference indicating when the IOG periods occur. Both of FIGS. **7A** and **7B** show original data along with smoothed data.

[0072] FIGS. **7A** and **7B** demonstrate that there is a strong correlation between implement motion features and the IOG periods. For example, FIG. **7A** shows that the IOG periods are strongly correlated with a decreasing bucket distance. In some embodiments, maximum and minimum values (i.e., peaks and valleys, respectively) of the bucket distance may be identified and the IOG period (including the IOG start and end times) may be constrained to reside between the times associated with the maximum and minimum values. As another example, FIG. **7B** shows that the IOG periods are strongly correlated with a decreasing bucket rotation. In some embodiments, maximum and minimum values of the bucket rotation may be identified and the IOG period (including the IOG start and end times) may be constrained to reside between the times associated with the maximum and minimum values.

[0073] FIG. **8** illustrates an example of a prediction of IOG periods based on a vibration signal and its corresponding amplitude features **847**, in accordance with some embodiments of the present disclosure. Also shown are predictions for IOG start times **836** and IOG end times **838** and the truth reference indicating when the IOG periods occur. In the illustrated example, amplitude features **847** represent the changes in the amplitude of the vibration/acceleration signal. Each value in amplitude features **847** is calculated by summing the magnitude of a first window including 8 values and dividing by the sum of the magnitude of a second window including 64 values. The two windows are aligned at the last sample of the windows.

[0074] The illustrated amplitude features of the vibration signal are provided to a machine-learning model which generates IOG candidates **844** and IIA candidates **845**, which are illustrated in FIG. **8** at their associated times. During a post-processing stage, clusters of IOG candidates **844** and clusters of IIA candidates **845** may be identified. The transitions from clusters of IIA candidates **845** to clusters of IOG candidates **844** may be used to predict IOG start times **836** based on, for example, the time of the first IOG candidate in each IOG cluster. The transitions from clusters of IOG candidates **844** to clusters of IIA candidates **845** may be used to predict IOG end times **838** based on, for example, the time of the first IIA candidate in each IIA cluster and/or the time of the last IOG candidate in each IOG cluster.

[0075] FIG. **9** illustrates an example of a prediction of an IOG end time **938** based on a vibration

signal **942** and an implement distance (bucket distance), in accordance with some embodiments of the present disclosure. Also shown is a truth reference indicating when the IOG periods occur. In the illustrated example, several IOG candidates are generated by a machine-learning model based on the features of the vibration signal. During a post-processing stage, a cluster of IOG candidates are identified and IOG start and end times are predicted. In some embodiments, the IOG start and end times are constrained to be between the maximum implement distance (maximum bucket distance) and minimum implement distance (minimum bucket distance).

[0076] In some embodiments, it may be determined whether the outlier IOG candidate (at roughly 2.4 minutes) should be used to predict IOG end time **838** or whether it should be disregarded. By consulting the time difference between the IOG candidate at 2.4 minutes and the time of the minimum bucket distance (also at roughly 2.4 minutes), it may be determined that the IOG candidate at 2.4 minutes should be used for predicting IOG end time **838** and not be disregarded. In some embodiments, the time difference between a particular IOG candidate and the time of the minimum bucket distance may be compared to a threshold to determine whether the particular IOG candidate should be considered to determine the IOG end time, where a time difference above the threshold causes the particular IOG candidate to be disregarded.

[0077] FIG. **10** illustrates an example method **1000**, in accordance with some embodiments of the present disclosure. In various embodiments, method **1000** may be considered to be a method of determining a period during which an implement of a construction machine is interacting with a ground surface, a method of determining a start time and/or end time of a period during which an implement of a construction machine is interacting with a ground surface, a method of adjusting a ground surface map, a method of generating a ground surface map, among other possibilities. One or more steps of method **1000** may be omitted during performance of method **1000**, and steps of method **1000** may be performed in any order and/or in parallel. One or more steps of method **1000** may be performed by one or more processors, such as those included in a control unit (e.g., control units **360**, **460**). Method **1000** may be implemented as a computer-readable medium or computer program product comprising instructions which, when the program is executed by one or more computers, cause the one or more computers to carry out the steps of method **1000**.

[0078] At step **1002**, a vibration signal (e.g., vibration signals **242**, **342**, **442**, **542**, **642**, **942**) is captured. The vibration signal may be captured using a vibration sensor (e.g., vibration sensors **365**, **465**). The vibration sensor may be mounted to an implement (e.g., implements **121**, **221**, **321**) of a construction machine (e.g., construction machines **150**, **350**). The vibration signal may be indicative of a movement of the implement. In some embodiments, the vibration sensor may include an accelerometer and the vibration signal may include an acceleration signal. In some embodiments, the vibration sensor may include a gyroscope and the vibration signal may include a rotation signal. In some embodiments, the vibration sensor may include an IMU sensor (such as an accelerometer or a gyroscope) and the vibration signal may include an IMU signal (such as an acceleration signal or a rotation signal). In some embodiments, the vibration sensor may include an acoustic sensor (such as a microphone).

[0079] At step **1004**, one or more features (e.g., features **547**, **647**, **847**) are extracted from the vibration signal. The one or more features may include signal amplitude features (e.g., signal amplitude features **547-1**), signal frequency features (e.g., signal frequency features **547-2**), implement motion features (e.g., implement motion features **547-3**), among other possibilities. The one or more features may be extracted by a feature extractor (e.g., feature extractor **546**). The feature extractor may extract the one or more features from a set of implement positions (e.g., implement positions **529**) in addition to the vibration signal.

[0080] At step **1006**, the one or more features are provided to a machine-learning model (e.g., machine-learning model **532**). Upon providing the one or more features to the machine-learning model, the machine-learning model may generate a model output based on the one or more features. The model output may include a set of IOG candidates (e.g., IOG candidates **544**, **844**)

and/or a set of IIA candidates (e.g., IIA candidates **545, 845**). The set of IOG candidates may correspond to times at which the implement is interacting with the ground surface. The set of IIA candidates may correspond to times that the implement is not interacting with the ground surface (i.e., the implement is above the ground surface). The machine-learning model may be a pre-trained support-vector machine, a neural network, among other possibilities.

[0081] At step **1008**, an IOG start time (e.g., IOG start times **336, 536, 836**) and/or an IOG end time (e.g., IOG end times **338, 538, 838, 938**) are predicted based on the model output. The IOG start time and the IOG end time may form an IOG period (e.g., IOG periods **340, 540**) during which the implement is interacting with a ground surface (e.g., ground surfaces **112, 212**) at a construction site (e.g., construction site **310**). The implement may be interacting with the ground surface when the implement is cutting or digging into the ground surface. The IOG start time and/or the IOG end time may be predicted based on the set of IOG candidates and/or the set of IIA candidates. In some embodiments, the IOG start time and/or the IOG end time may be predicted based on the implement motion features and/or the set of implement positions.

[0082] At step **1010**, a ground surface map (e.g., ground surface map **124**) is adjusted based on the period (and/or the IOG start time and the IOG end time) during which the implement is interacting with the ground surface. The ground surface map may be adjusted further based on a path of the implement during the period. Adjusting the ground surface map may include modifying/changing/updating the ground surface map to reflect the path of the implement during the period. In some embodiments, the ground surface map may be set equal to the path of the implement (e.g., the positions of the bucket edge) during the period.

[0083] It should be noted that method **1000** may be modified to predict different periods associated with operation of the construction machine in addition to or alternative to predicting the IOG period. For example, in various embodiments, method **1000** may be modified to predict a time period in which the construction machine is dumping, slewing, tramming, idling, reversing, accelerating, decelerating, among other possibilities.

[0084] FIG. **11** illustrates an example computer system **1100** comprising various hardware elements, in accordance with some embodiments of the present disclosure. Computer system **1100** may be incorporated into or integrated with devices described herein and/or may be configured to perform some or all of the steps of the methods provided by various embodiments. For example, in various embodiments, computer system **1100** may be incorporated into machine control system **400** and/or may be configured to perform method **1000**. It should be noted that FIG. **11** is meant only to provide a generalized illustration of various components, any or all of which may be utilized as appropriate. FIG. **11**, therefore, broadly illustrates how individual system elements may be implemented in a relatively separated or relatively more integrated manner.

[0085] In the illustrated example, computer system **1100** includes a communication medium **1102**, one or more processor(s) **1104**, one or more input device(s) **1106**, one or more output device(s) **1108**, a communications subsystem **1110**, and one or more memory device(s) **1112**. Computer system **1100** may be implemented using various hardware implementations and embedded system technologies. For example, one or more elements of computer system **1100** may be implemented as a field-programmable gate array (FPGA), such as those commercially available by XILINX®, INTEL®, or LATTICE SEMICONDUCTOR®, a system-on-a-chip (SoC), an application-specific integrated circuit (ASIC), an application-specific standard product (ASSP), a microcontroller, and/or a hybrid device, such as an SoC FPGA, among other possibilities.

[0086] The various hardware elements of computer system **1100** may be communicatively coupled via communication medium **1102**. While communication medium **1102** is illustrated as a single connection for purposes of clarity, it should be understood that communication medium **1102** may include various numbers and types of communication media for transferring data between hardware elements. For example, communication medium **1102** may include one or more wires (e.g., conductive traces, paths, or leads on a printed circuit board (PCB) or integrated circuit (IC),

microstrips, striplines, coaxial cables), one or more optical waveguides (e.g., optical fibers, strip waveguides), and/or one or more wireless connections or links (e.g., infrared wireless communication, radio communication, microwave wireless communication), among other possibilities.

[0087] In some embodiments, communication medium **1102** may include one or more buses connecting pins of the hardware elements of computer system **1100**. For example, communication medium **1102** may include a bus that connects processor(s) **1104** with main memory **1114**, referred to as a system bus, and a bus that connects main memory **1114** with input device(s) **1106** or output device(s) **1108**, referred to as an expansion bus. The system bus may itself consist of several buses, including an address bus, a data bus, and a control bus. The address bus may carry a memory address from processor(s) **1104** to the address bus circuitry associated with main memory **1114** in order for the data bus to access and carry the data contained at the memory address back to processor(s) **1104**. The control bus may carry commands from processor(s) **1104** and return status signals from main memory **1114**. Each bus may include multiple wires for carrying multiple bits of information and each bus may support serial or parallel transmission of data.

[0088] Processor(s) **1104** may include one or more central processing units (CPUs), graphics processing units (GPUs), neural network processors or accelerators, digital signal processors (DSPs), and/or other general-purpose or special-purpose processors capable of executing instructions. A CPU may take the form of a microprocessor, which may be fabricated on a single IC chip of metal-oxide-semiconductor field-effect transistor (MOSFET) construction. Processor(s) **1104** may include one or more multi-core processors, in which each core may read and execute program instructions concurrently with the other cores, increasing speed for programs that support multithreading.

[0089] Input device(s) **1106** may include one or more of various user input devices such as a mouse, a keyboard, a microphone, as well as various sensor input devices, such as an image capture device, a pressure sensor (e.g., barometer, tactile sensor), a temperature sensor (e.g., thermometer, thermocouple, thermistor), a movement sensor (e.g., accelerometer, gyroscope, tilt sensor), a light sensor (e.g., photodiode, photodetector, charge-coupled device), and/or the like. Input device(s) **1106** may also include devices for reading and/or receiving removable storage devices or other removable media. Such removable media may include optical discs (e.g., Blu-ray discs, DVDs, CDs), memory cards (e.g., CompactFlash card, Secure Digital (SD) card, Memory Stick), floppy disks, Universal Serial Bus (USB) flash drives, external hard disk drives (HDDs) or solid-state drives (SSDs), and/or the like.

[0090] Output device(s) **1108** may include one or more of various devices that convert information into human-readable form, such as without limitation a display device, a speaker, a printer, a haptic or tactile device, and/or the like. Output device(s) **1108** may also include devices for writing to removable storage devices or other removable media, such as those described in reference to input device(s) **1106**. Output device(s) **1108** may also include various actuators for causing physical movement of one or more components. Such actuators may be hydraulic, pneumatic, electric, and may be controlled using control signals generated by computer system **1100**.

[0091] Communications subsystem **1110** may include hardware components for connecting computer system **1100** to systems or devices that are located external to computer system **1100**, such as over a computer network. In various embodiments, communications subsystem **1110** may include a wired communication device coupled to one or more input/output ports (e.g., a universal asynchronous receiver-transmitter (UART)), an optical communication device (e.g., an optical modem), an infrared communication device, a radio communication device (e.g., a wireless network interface controller, a BLUETOOTH® device, an IEEE 802.11 device, a Wi-Fi device, a Wi-Max device, a cellular device), among other possibilities.

[0092] Memory device(s) **1112** may include the various data storage devices of computer system **1100**. For example, memory device(s) **1112** may include various types of computer memory with

various response times and capacities, from faster response times and lower capacity memory, such as processor registers and caches (e.g., L0, L1, L2), to medium response time and medium capacity memory, such as random-access memory (RAM), to lower response times and lower capacity memory, such as solid-state drives and hard drive disks. While processor(s) **1104** and memory device(s) **1112** are illustrated as being separate elements, it should be understood that processor(s) **1104** may include varying levels of on-processor memory, such as processor registers and caches that may be utilized by a single processor or shared between multiple processors.

[0093] Memory device(s) **1112** may include main memory **1114**, which may be directly accessible by processor(s) **1104** via the memory bus of communication medium **1102**. For example, processor(s) **1104** may continuously read and execute instructions stored in main memory **1114**. As such, various software elements may be loaded into main memory **1114** to be read and executed by processor(s) **1104** as illustrated in FIG. **11**. Typically, main memory **1114** is volatile memory, which loses all data when power is turned off and accordingly needs power to preserve stored data. Main memory **1114** may further include a small portion of non-volatile memory containing software (e.g., firmware, such as BIOS) that is used for reading other software stored in memory device(s) **1112** into main memory **1114**. In some embodiments, the volatile memory of main memory **1114** is implemented as RAM, such as dynamic random-access memory (DRAM), and the non-volatile memory of main memory **1114** is implemented as read-only memory (ROM), such as flash memory, erasable programmable read-only memory (EPROM), or electrically erasable programmable read-only memory (EEPROM).

[0094] Computer system **1100** may include software elements, shown as being currently located within main memory **1114**, which may include an operating system, device driver(s), firmware, compilers, and/or other code, such as one or more application programs, which may include computer programs provided by various embodiments of the present disclosure. Merely by way of example, one or more steps described with respect to any methods discussed above, may be implemented as instructions **1116**, which are executable by computer system **1100**. In one example, such instructions **1116** may be received by computer system **1100** using communications subsystem **1110** (e.g., via a wireless or wired signal that carries instructions **1116**), carried by communication medium **1102** to memory device(s) **1112**, stored within memory device(s) **1112**, read into main memory **1114**, and executed by processor(s) **1104** to perform one or more steps of the described methods. In another example, instructions **1116** may be received by computer system **1100** using input device(s) **1106** (e.g., via a reader for removable media), carried by communication medium **1102** to memory device(s) **1112**, stored within memory device(s) **1112**, read into main memory **1114**, and executed by processor(s) **1104** to perform one or more steps of the described methods.

[0095] In some embodiments of the present disclosure, instructions **1116** are stored on a computer-readable storage medium (or simply computer-readable medium). Such a computer-readable medium may be non-transitory and may therefore be referred to as a non-transitory computer-readable medium. In some cases, the non-transitory computer-readable medium may be incorporated within computer system **1100**. For example, the non-transitory computer-readable medium may be one of memory device(s) **1112** (as shown in FIG. **11**). In some cases, the non-transitory computer-readable medium may be separate from computer system **1100**. In one example, the non-transitory computer-readable medium may be a removable medium provided to input device(s) **1106** (as shown in FIG. **11**), such as those described in reference to input device(s) **1106**, with instructions **1116** being read into computer system **1100** by input device(s) **1106**. In another example, the non-transitory computer-readable medium may be a component of a remote electronic device, such as a mobile phone, that may wirelessly transmit a data signal that carries instructions **1116** to computer system **1100** and that is received by communications subsystem **1110** (as shown in FIG. **11**).

[0096] Instructions **1116** may take any suitable form to be read and/or executed by computer system **1100**. For example, instructions **1116** may be source code (written in a human-readable

programming language such as Java, C, C++, C#, Python), object code, assembly language, machine code, microcode, executable code, and/or the like. In one example, instructions **1116** are provided to computer system **1100** in the form of source code, and a compiler is used to translate instructions **1116** from source code to machine code, which may then be read into main memory **1114** for execution by processor(s) **1104**. As another example, instructions **1116** are provided to computer system **1100** in the form of an executable file with machine code that may immediately be read into main memory **1114** for execution by processor(s) **1104**. In various examples, instructions **1116** may be provided to computer system **1100** in encrypted or unencrypted form, compressed or uncompressed form, as an installation package or an initialization for a broader software deployment, among other possibilities.

[0097] In one aspect of the present disclosure, a system (e.g., computer system **1100**) is provided to perform methods in accordance with various embodiments of the present disclosure. For example, some embodiments may include a system comprising one or more processors (e.g., processor(s) **1104**) that are communicatively coupled to a non-transitory computer-readable medium (e.g., memory device(s) **1112** or main memory **1114**). The non-transitory computer-readable medium may have instructions (e.g., instructions **1116**) stored therein that, when executed by the one or more processors, cause the one or more processors to perform the methods described in the various embodiments.

[0098] In another aspect of the present disclosure, a computer-program product that includes instructions (e.g., instructions **1116**) is provided to perform methods in accordance with various embodiments of the present disclosure. The computer-program product may be tangibly embodied in a non-transitory computer-readable medium (e.g., memory device(s) **1112** or main memory **1114**). The instructions may be configured to cause one or more processors (e.g., processor(s) **1104**) to perform the methods described in the various embodiments.

[0099] In another aspect of the present disclosure, a non-transitory computer-readable medium (e.g., memory device(s) **1112** or main memory **1114**) is provided. The non-transitory computer-readable medium may have instructions (e.g., instructions **1116**) stored therein that, when executed by one or more processors (e.g., processor(s) **1104**), cause the one or more processors to perform the methods described in the various embodiments.

[0100] The methods, systems, and devices discussed above are examples. Various configurations may omit, substitute, or add various procedures or components as appropriate. For instance, in alternative configurations, the methods may be performed in an order different from that described, and/or various stages may be added, omitted, and/or combined. Also, features described with respect to certain configurations may be combined in various other configurations. Different aspects and elements of the configurations may be combined in a similar manner. Also, technology evolves and, thus, many of the elements are examples and do not limit the scope of the disclosure or claims.

[0101] Specific details are given in the description to provide a thorough understanding of exemplary configurations including implementations. However, configurations may be practiced without these specific details. For example, well-known circuits, processes, algorithms, structures, and techniques have been shown without unnecessary detail in order to avoid obscuring the configurations. This description provides example configurations only, and does not limit the scope, applicability, or configurations of the claims. Rather, the preceding description of the configurations will provide those skilled in the art with an enabling description for implementing described techniques. Various changes may be made in the function and arrangement of elements without departing from the spirit or scope of the disclosure.

[0102] Having described several example configurations, various modifications, alternative constructions, and equivalents may be used without departing from the spirit of the disclosure. For example, the above elements may be components of a larger system, wherein other rules may take precedence over or otherwise modify the application of the technology. Also, a number of steps

may be undertaken before, during, or after the above elements are considered. Accordingly, the above description does not bind the scope of the claims.

[0103] As used herein and in the appended claims, the singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise. Thus, for example, reference to “a user” includes reference to one or more of such users, and reference to “a processor” includes reference to one or more processors and equivalents thereof known to those skilled in the art, and so forth.

[0104] Also, the words “comprise,” “comprising,” “contains,” “containing,” “include,” “including,” and “includes,” when used in this specification and in the following claims, are intended to specify the presence of stated features, integers, components, or steps, but they do not preclude the presence or addition of one or more other features, integers, components, steps, acts, or groups.

[0105] It is also understood that the examples and embodiments described herein are for illustrative purposes only and that various modifications or changes in light thereof will be suggested to persons skilled in the art and are to be included within the spirit and purview of this application and scope of the appended claims.

Claims

1. A construction machine comprising: a vehicle body; an engine for enabling movement of the construction machine, the engine coupled to the vehicle body; an implement coupled to the vehicle body; a vibration sensor mounted to the implement, wherein the vibration sensor is configured to capture a vibration signal that is indicative of a movement of the implement; one or more processors configured to perform operations for determining a period **8** during which the implement is interacting with a ground surface, wherein the one or more processors are configured to: receive the vibration signal from the vibration sensor; extract one or more features from the vibration signal; generate a model output based on the one or more features using a machine-learning model, the model output including a set of implement-on-ground (IOG) candidates corresponding to times at which the implement is interacting with the ground surface and a set of implement-in-air (IIA) candidates corresponding to times at which the implement is not interacting with the ground surface; and predict an IOG start time based on a transition between a first cluster of the set of IIA candidates and a cluster of the set of IOG candidates and an IOG end time based on a transition between the cluster of the set of IOG candidates and a second cluster of the set of IIA candidates, the IOG start time and the IOG end time forming the period during which the implement is interacting with the ground surface.
2. The construction machine of claim 1, wherein the construction machine is an excavator.
3. The construction machine of claim 1, wherein the construction machine is a grader.
4. The construction machine of claim 1, wherein the construction machine is a bulldozer.
5. The construction machine of claim 1, wherein the construction machine is a backhoe.
6. The construction machine of claim 1, wherein the vibration sensor includes an accelerometer and the vibration signal includes an acceleration signal.
7. The construction machine of claim 1, wherein the vibration sensor includes a gyroscope and the vibration signal includes a rotation signal.
8. The construction machine of claim 1, wherein the one or more features include at least one of signal amplitude features or signal frequency features.
9. The construction machine of claim 1, wherein the machine-learning model is a pre-trained support-vector machine.
10. The construction machine of claim 1, wherein the machine-learning model is a pre-trained neural network.
11. The construction machine of claim 1, wherein the machine-learning model is a pre-trained convolutional neural network.

12. A construction machine comprising: a vehicle body; an implement coupled to the vehicle body; and a machine control system comprising: a vibration sensor mounted to the implement, wherein the vibration sensor is configured to capture a vibration signal that is indicative of a movement of the implement; a control unit configured to perform operations for determining a period during which the implement is interacting with a ground surface, wherein the control unit is configured to: receive the vibration signal from the vibration sensor; extract one or more features from the vibration signal; generate a model output based on the one or more features using a machine-learning model, the model output including a set of implement-on-ground (IOG) candidates corresponding to times at which the implement is interacting with the ground surface and a set of implement-in-air (IIA) candidates corresponding to times at which the implement is not interacting with the ground surface; and predict an IOG start time based on a transition between a first cluster of the set of IIA candidates and a cluster of the set of IOG candidates and an IOG end time based on a transition between the cluster of the set of IOG candidates and a second cluster of the set of IIA candidates, the IOG start time and the IOG end time forming the period during which the implement is interacting with the ground surface.

13. The construction machine of claim 12, wherein the vibration sensor includes an accelerometer and the vibration signal includes an acceleration signal.

14. The construction machine of claim 12, wherein the vibration sensor includes a gyroscope and the vibration signal includes a rotation signal.

15. The construction machine of claim 12, wherein the one or more features include at least one of signal amplitude features or signal frequency features.

16. The construction machine of claim 12, wherein the machine-learning model is a pre-trained support-vector machine.

17. A machine control system integrated with a construction machine, the machine control system comprising: a vibration sensor mounted to an implement of the construction machine, wherein the vibration sensor is configured to capture a vibration signal that is indicative of a movement of the implement; a control unit configured to perform operations for determining a period during which the implement is interacting with a ground surface, wherein the control unit is configured to: receive the vibration signal from the vibration sensor; extract one or more features from the vibration signal; generate a model output based on the one or more features using a machine-learning model, the model output including a set of implement-on-ground (IOG) candidates corresponding to times at which the implement is interacting with the ground surface and a set of implement-in-air (IIA) candidates corresponding to times at which the implement is not interacting with the ground surface; and predict an IOG start time based on a transition between a first cluster of the set of IIA candidates and a cluster of the set of IOG candidates and an IOG end time based on a transition between the cluster of the set of IOG candidates and a second cluster of the set of IIA candidates, the IOG start time and the IOG end time forming the period during which the implement is interacting with the ground surface.

18. The machine control system of claim 17, wherein the vibration sensor includes an accelerometer and the vibration signal includes an acceleration signal.

19. The machine control system of claim 17, wherein the vibration sensor includes a gyroscope and the vibration signal includes a rotation signal.

20. The machine control system of claim 17, wherein the one or more features include at least one of signal amplitude features or signal frequency features.
